

1 **Theoretical uncertainties in the extraction of CKM angles**
2 **from $b \rightarrow c\bar{c}s$ transitions**

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Abstract

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I. INTRODUCTION

II. DECAY AMPLITUDES

We divide the decay channels into three classes of $SU(3)$ -related modes:

- Class I: $\mathcal{B} \rightarrow J/\psi P_8$, with $\mathcal{B} = B^+, B_d, B_s$ and P_8 a member of the light pseudoscalar meson octet;

- Class II: $\mathcal{B} \rightarrow J/\psi V_8$, with V_8 a member of the light vector meson octet;

- Class III: $\mathcal{B} \rightarrow \mathcal{D}\bar{\mathcal{D}}$, with $\mathcal{D} = D^+, D^0, D_s^+$.

Furthermore, due to the presence of η - η' and ϕ - ω mixing, some decay channels receive contributions also from $SU(3)$ singlet mesons (P_1 and V_1), so that Class I is supplemented by the channels $\mathcal{B} \rightarrow J/\psi P_1$ and Class II by the channels $\mathcal{B} \rightarrow J/\psi V_1$.

Let us first write down the amplitudes in the isospin limit, neglecting electroweak penguin contributions, grouping the various contributions according to their CKM factors. Denoting

so $q = u, d$, for Class I we have

$$\mathcal{A}(B^+ \rightarrow \pi^+ J/\Psi) = \lambda_{bd}^c \widetilde{E}_2(c, c, q, B, J/\Psi, \pi) + \quad (1)$$

$$\lambda_{bd}^u (\text{EA}_1(q, q, c, B, \pi, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(q, c, B, J/\Psi, \pi)),$$

$$\mathcal{A}(B^+ \rightarrow K^+ J/\Psi) = \lambda_{bs}^c \widetilde{E}_2(c, c, s, B, J/\Psi, K) + \quad (2)$$

$$\lambda_{bs}^u (\text{EA}_1(s, q, c, B, K, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(s, c, B, J/\Psi, K)),$$

$$\sqrt{2}\mathcal{A}(B_d \rightarrow \pi^0 J/\Psi) = \lambda_{bd}^c \widetilde{E}_2(c, c, q, B, J/\Psi, \pi) - \quad (3)$$

$$\lambda_{bd}^u (\text{EA}_2(q, q, c, B, \pi, J/\Psi) + \widetilde{P}_2^{\text{GIM}}(q, c, B, J/\Psi, \pi)),$$

$$\mathcal{A}(B_d \rightarrow K^0 J/\Psi) = \lambda_{bs}^c \widetilde{E}_2(c, c, s, B, J/\Psi, K) - \quad (4)$$

$$\lambda_{bs}^u \widetilde{P}_2^{\text{GIM}}(s, c, B, J/\Psi, K),$$

$$\sqrt{3}\mathcal{A}(B_d \rightarrow \eta_1 J/\Psi) = \lambda_{bd}^c (\widetilde{E}_2(c, c, q, B, J/\Psi, \eta_1) + \quad (5)$$

$$\widetilde{E}\widetilde{A}_2(c, c, s, B, J/\Psi, \eta_1) + 2\widetilde{E}\widetilde{A}_2(c, c, q, B, J/\Psi, \eta_1)) +$$

$$\lambda_{bd}^u (\text{EA}_2(q, q, c, B, \eta_1, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(q, c, B, J/\Psi, \eta_1) -$$

$$\widetilde{P}_4^{\text{GIM}}(c, s, B, J/\Psi, \eta_1) - 2\widetilde{P}_4^{\text{GIM}}(q, c, B, \eta_1, J/\Psi)),$$

$$\sqrt{6}\mathcal{A}(B_d \rightarrow \eta_8 J/\Psi) = \lambda_{bd}^c (\widetilde{E}_2(c, c, q, B, J/\Psi, \eta_8) - \quad (6)$$

$$2\widetilde{E}\widetilde{A}_2(c, c, s, B, J/\Psi, \eta_8) + 2\widetilde{E}\widetilde{A}_2(c, c, q, B, J/\Psi, \eta_8)) +$$

$$\lambda_{bd}^u (\text{EA}_2(q, q, c, B, \eta_8, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(q, c, B, J/\Psi, \eta_8) +$$

$$2\widetilde{P}_4^{\text{GIM}}(c, s, B, J/\Psi, \eta_8) - 2\widetilde{P}_4^{\text{GIM}}(c, q, B, J/\Psi, \eta_8)),$$

$$\sqrt{2}\mathcal{A}(B_s \rightarrow \pi^0 J/\Psi) = -\lambda_{bs}^u \text{EA}_2(q, q, c, B_s, \pi, J/\Psi), \quad (7)$$

$$\mathcal{A}(B_s \rightarrow \bar{K}^0 J/\Psi) = \lambda_{bd}^c \widetilde{E}_2(c, c, q, B_s, J/\Psi, \bar{K}) - \quad (8)$$

$$\lambda_{bd}^u \widetilde{P}_2^{\text{GIM}}(q, c, B_s, J/\Psi, \bar{K}),$$

$$\sqrt{3}\mathcal{A}(B_s \rightarrow \eta_1 J/\Psi) = \lambda_{bs}^c (\widetilde{E}_2(c, c, s, B_s, J/\Psi, \eta_1) + \quad (9)$$

$$\widetilde{E}\widetilde{A}_2(c, c, s, B_s, J/\Psi, \eta_1) + 2\widetilde{E}\widetilde{A}_2(c, c, q, B_s, J/\Psi, \eta_1)) +$$

$$\lambda_{bs}^u (\text{EA}_2(q, q, c, B_s, \eta_1, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(s, c, B_s, J/\Psi, \eta_1) -$$

$$\widetilde{P}_4^{\text{GIM}}(c, s, B_s, J/\Psi, \eta_1) - 2\widetilde{P}_4^{\text{GIM}}(c, q, B_s, J/\Psi, \eta_1)),$$

$$\sqrt{6}\mathcal{A}(B_s \rightarrow \eta_8 J/\Psi) = -2\lambda_{bs}^c (\widetilde{E}_2(c, c, s, B_s, J/\Psi, \eta_8) + \quad (10)$$

$$\widetilde{E}\widetilde{A}_2(c, c, s, B_s, J/\Psi, \eta_8) - \widetilde{E}\widetilde{A}_2(c, c, q, B_s, J/\Psi, \eta_8)) +$$

$$\lambda_{bs}^u (\text{EA}_2(q, q, c, B_s, \eta_8, J/\Psi) + 2\widetilde{P}_2^{\text{GIM}}(s, c, B_s, J/\Psi, \eta_8) +$$

$$2\widetilde{P}_4^{\text{GIM}}(c, s, B_s, J/\Psi, \eta_8) - 2\widetilde{P}_4^{\text{GIM}}(c, q, B_s, J/Psi, \eta_8)),$$

where $\lambda_{q_1 q_2}^{q_3} = V_{q_3 q_1} V_{q_3 q_2}^*$ and the various functions represent different topologies contributing to the decay amplitudes, in the notation of Ref. [1], with the following additional definitions:

$$\widetilde{E}_2(c, c, x, \mathcal{B}, J/\Psi, P_8) \equiv E_2(c, c, x, \mathcal{B}, J/\Psi, P_8) + P_2(x, c, \mathcal{B}, J/\Psi, P_8), \quad (11)$$

$$\widetilde{P}_2^{\text{GIM}}(x, c, \mathcal{B}, J/\Psi, P_8) \equiv P_2^{\text{GIM}}(x, c, \mathcal{B}, J/\Psi, P_8) - P_2(x, c, \mathcal{B}, J/\Psi, P_8), \quad (12)$$

$$\widetilde{EA}_2(c, c, x, \mathcal{B}, J/\Psi, P_8) \equiv EA_2(c, c, x, \mathcal{B}, J/\Psi, P_8) + P_4(c, x, \mathcal{B}, J/\Psi, P_8), \quad (13)$$

$$\widetilde{P}_4^{\text{GIM}}(c, x, \mathcal{B}, J/\Psi, P_8) \equiv P_4^{\text{GIM}}(c, x, \mathcal{B}, J/\Psi, P_8) - P_4(c, x, \mathcal{B}, J/\Psi, P_8). \quad (14)$$

For Class II decays we have

$$\mathcal{A}(B^+ \rightarrow \rho^+ J/\Psi) = \lambda_{bd}^c \widetilde{E}_2(c, c, q, B, J/\Psi, \rho) + \quad (15)$$

$$\lambda_{bd}^u (EA_1(q, q, c, B, \rho, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(q, c, B, J/\Psi, \rho)),$$

$$\mathcal{A}(B^+ \rightarrow K^{*+} J/\Psi) = \lambda_{bs}^c \widetilde{E}_2(c, c, s, B, J/\Psi, K^*) + \quad (16)$$

$$\lambda_{bs}^u (EA_1(s, q, c, B, K^*, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(s, c, B, J/\Psi, K^*)),$$

$$\sqrt{2}\mathcal{A}(B_d \rightarrow \rho^0 J/\Psi) = \lambda_{bd}^c \widetilde{E}_2(c, c, q, B, J/\Psi, \rho) - \quad (17)$$

$$\lambda_{bd}^u (EA_2(q, q, c, B, \rho, J/\Psi) + \widetilde{P}_2^{\text{GIM}}(q, c, B, J/\Psi, \rho)),$$

$$\mathcal{A}(B_d \rightarrow K^{*0} J/\Psi) = \lambda_{bs}^c \widetilde{E}_2(c, c, s, B, J/\Psi, K^*) -$$

$$\lambda_{bs}^u \widetilde{P}_2^{\text{GIM}}(s, c, B, J/\Psi, K^*), \quad (18)$$

$$\mathcal{A}(B_d \rightarrow \phi J/\Psi) = -\lambda_{bd}^c \widetilde{EA}_2(c, c, s, B, J/\Psi, \phi) + \quad (19)$$

$$\lambda_{bd}^u \widetilde{P}_4^{\text{GIM}}(c, s, B, J/\Psi, \phi),$$

$$\sqrt{2}\mathcal{A}(B_d \rightarrow \omega J/\Psi) = \lambda_{bd}^c (\widetilde{E}_2(c, c, q, B, J/\Psi, \omega) + 2\widetilde{EA}_2(c, c, q, B, J/\Psi, \omega)) + \quad (20)$$

$$\lambda_{bd}^u (EA_2(q, q, c, B, \omega, J/\Psi) - \widetilde{P}_2^{\text{GIM}}(q, c, B, J/\Psi, \omega) -$$

$$2\widetilde{P}_4^{\text{GIM}}(c, q, B, J/\Psi, \omega)),$$

$$\sqrt{2}\mathcal{A}(B_s \rightarrow \rho^0 J/\Psi) = -\lambda_{bs}^u EA_2(q, q, c, B_s, R, J/\Psi), \quad (21)$$

$$\mathcal{A}(B_s \rightarrow \bar{K}^{*0} J/\Psi) = \lambda_{bd}^c \widetilde{E}_2(c, c, q, B_s, J/\Psi, \bar{K}^*) - \quad (22)$$

$$\lambda_{bd}^u \widetilde{P}_2^{\text{GIM}}(q, c, B_s, J/\Psi, \bar{K}^*),$$

$$\mathcal{A}(B_s \rightarrow \phi J/\Psi) = -\lambda_{bs}^c (\widetilde{E}_2(c, c, s, B_s, J/\Psi, \phi) + \widetilde{EA}_2(c, c, s, B_s, J/\Psi, \phi)) + \quad (23)$$

$$\lambda_{bs}^u (\widetilde{P}_2^{\text{GIM}}(s, c, B_s, J/\Psi, \phi) + \widetilde{P}_4^{\text{GIM}}(c, s, B_s, J/\Psi, \phi)),$$

$$\sqrt{2}\mathcal{A}(B_s \rightarrow \omega J/\Psi) = 2\lambda_{bs}^c \widetilde{EA}_2(c, c, q, B_s, J/\Psi, \omega) + \quad (24)$$

$$\lambda_{bs}^u (EA_2(q, q, c, B_s, \omega, J/\Psi) - 2\widetilde{P}_4^{\text{GIM}}(c, q, B_s, J/\Psi, \omega)),$$

34 where we have assumed ideal mixing for the ω and ϕ mesons. For Class III decays we have

$$\mathcal{A}(B^+ \rightarrow D^+ \bar{D}^0) = \lambda_{bd}^c \tilde{E}_1(q, c, c, B, D, \bar{D}) + \quad (25)$$

$$\lambda_{bd}^u (A_1(q, c, q, B, D, \bar{D}) - \tilde{P}_1^{\text{GIM}}(q, c, B, D, \bar{D})),$$

$$\mathcal{A}(B^+ \rightarrow D_s^+ \bar{D}^0) = \lambda_{bs}^c \tilde{E}_1(s, c, c, B, D_s^+, \bar{D}) + \quad (26)$$

$$\lambda_{bs}^u (A_1(s, c, q, B, D_s^+, \bar{D}) - \tilde{P}_1^{\text{GIM}}(s, c, B, D_s^+, \bar{D})),$$

$$\mathcal{A}(B_d \rightarrow D^+ D^-) = \lambda_{bd}^c (\tilde{E}_1(q, c, c, B, D, \bar{D}) + \tilde{A}_2(c, q, c, B, \bar{D}, D)) - \quad (27)$$

$$\lambda_{bd}^u (\tilde{P}_1^{\text{GIM}}(q, c, B, D, \bar{D}) + \tilde{P}_3^{\text{GIM}}(c, q, B, D, \bar{D})),$$

$$\mathcal{A}(B_d \rightarrow D_s^+ D^-) = \lambda_{bs}^c \tilde{E}_1(s, c, c, B, D_s^+, \bar{D}) - \quad (28)$$

$$\lambda_{bs}^u \tilde{P}_1^{\text{GIM}}(s, c, B, D_s^+, \bar{D}),$$

$$\mathcal{A}(B_d \rightarrow D_s^+ D_s^-) = \lambda_{bd}^c \tilde{A}_2(c, s, c, B, D_s^-, D_s^+) - \quad (29)$$

$$\lambda_{bd}^u \tilde{P}_3^{\text{GIM}}(c, s, B, D_s^+, D_s^-),$$

$$\mathcal{A}(B_d \rightarrow D^0 \bar{D}^0) = -\lambda_{bd}^c \tilde{A}_2(c, q, c, B, \bar{D}, D) - \quad (30)$$

$$\lambda_{bd}^u (A_2(q, c, q, B, D, \bar{D}) + \tilde{P}_3^{\text{GIM}}(q, c, B, \bar{D}, D)),$$

$$\mathcal{A}(B_s \rightarrow D^+ D^-) = \lambda_{bs}^c \tilde{A}_2(c, q, c, B_s, \bar{D}, D) - \quad (31)$$

$$\lambda_{bs}^u \tilde{P}_3^{\text{GIM}}(q, c, B_s, \bar{D}, D),$$

$$\mathcal{A}(B_s \rightarrow D^+ D_s^-) = \lambda_{bd}^c \tilde{E}_1(q, c, c, B_s, D, D_s^-) - \quad (32)$$

$$\lambda_{bd}^u \tilde{P}_1^{\text{GIM}}(q, c, B_s, D, D_s^-),$$

$$\mathcal{A}(B_s \rightarrow D_s^+ D_s^-) = \lambda_{bs}^c (\tilde{E}_1(s, c, c, B_s, D_s^+, D_s^-) + \tilde{A}_2(c, s, c, B_s, D_s^-, D_s^+)) - \quad (33)$$

$$\lambda_{bs}^u (\tilde{P}_1^{\text{GIM}}(s, c, B_s, D_s^+, D_s^-) + \tilde{P}_3^{\text{GIM}}(c, s, B_s, D_s^+, D_s^-)),$$

$$\mathcal{A}(B_s \rightarrow D^0 \bar{D}^0) = -\lambda_{bs}^c \tilde{A}_2(c, q, c, B_s, \bar{D}, D) - \quad (34)$$

$$\lambda_{bs}^u (A_2(q, c, q, B_s, D, \bar{D}) + \tilde{P}_3^{\text{GIM}}(q, c, B_s, \bar{D}, D)),$$

35 with the additional definitions

$$\tilde{E}_1(x, c, c, \mathcal{B}, \mathcal{D}, \bar{\mathcal{D}}) \equiv E_1(x, c, c, \mathcal{B}, \mathcal{D}, \bar{\mathcal{D}}) + P_1(x, c, \mathcal{B}, \mathcal{D}, \bar{\mathcal{D}}), \quad (35)$$

$$\tilde{P}_1^{\text{GIM}}(x, c, \mathcal{B}, \mathcal{D}, \bar{\mathcal{D}}) \equiv P_1^{\text{GIM}}(x, c, \mathcal{B}, \mathcal{D}, \bar{\mathcal{D}}) - P_1(x, c, \mathcal{B}, \mathcal{D}, \bar{\mathcal{D}}), \quad (36)$$

$$\tilde{A}_2(c, x, c, \mathcal{B}, \bar{\mathcal{D}}, \mathcal{D}) \equiv A_2(c, x, c, \mathcal{B}, \bar{\mathcal{D}}, \mathcal{D}) + P_3(c, x, \mathcal{B}, \bar{\mathcal{D}}, \mathcal{D}), \quad (37)$$

$$\tilde{P}_3^{\text{GIM}}(c, x, \mathcal{B}, \bar{\mathcal{D}}, \mathcal{D}) \equiv P_3^{\text{GIM}}(c, x, \mathcal{B}, \bar{\mathcal{D}}, \mathcal{D}) - P_3(c, x, \mathcal{B}, \bar{\mathcal{D}}, \mathcal{D}). \quad (38)$$

36 We now expand the above amplitudes around the $SU(3)$ limit, including a random com-
37 plex $SU(3)$ -breaking factor r for each U -spin transformation needed to relate the various

38 amplitudes to their $SU(3)$ limit. Concretely, this corresponds to multiplying each contribu-
 39 tion to the amplitudes above by a factor $(1+r_1+r_2+\dots+r_n)$, where n is the number of U -spin
 40 transformations needed to relate each term to the corresponding one in the $SU(3)$ limit. For
 41 instance, with respect to $\tilde{E}_2(c, c, q, B, J/\Psi, \pi)$ one has $n = 1$ for $\tilde{E}_2(c, c, s, B, J/\Psi, K)$ and
 42 for $\tilde{E}_2(c, c, q, B_s, J/\Psi, K)$, while $n = 2$ for $\tilde{E}_2(c, c, s, B_s, J/\Psi, \eta_8)$.

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